

Internship Progress Report: Ultra-Coherent Nanomechanical Resonators

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Abstract

This report details experimental and theoretical work on ultra-coherent nanomechanical resonators, including cryogenic characterization, spectral analysis of mechanical dissipation, and thermomechanical noise measurements. Key achievements include Voigt profile fitting of mechanical spectra, damping characterization under varied conditions, and further work focuses on reflection configuration thermomechanics.

1 Introduction

Nanomechanics is a field at the intersection of condensed matter physics, materials science, and quantum engineering, focused on understanding and manipulating the mechanical behavior of structures at the nanometer scale. These nanomechanical systems—commonly referred to as NEMS (nano-electromechanical systems)—exhibit unique vibrational modes, low effective mass, and high mechanical quality factors, making them ideal platforms for precision sensing and quantum-limited measurements.

One particularly impactful application of nanomechanics lies in the domain of cavity optomechanics, where a nanomechanical resonator is coupled to an electromagnetic cavity. The interplay between mechanical motion and confined electromagnetic fields allows for sensitive readout and manipulation of the resonator’s dynamics. Such systems have enabled real-time monitoring of thermomechanical motion, exploration of fundamental statistical physics at the mesoscopic scale, and control of mechanical degrees of freedom via radiation-pressure forces.

To extend our analysis of nanomechanical resonators fabricated by the Kippenberg group at EPFL, we are transitioning from a transmission-based measurement configuration to a reflection setup. The motivation for this shift stems from both practical and theoretical considerations. Reflection measurements enable the use of matched, open, or short terminations, which significantly reduce the effective electrical length of the feed-line and thereby help avoid spurious resonances that arise from standing waves in long transmission paths. In the case of an open termination, the setup also allows us to apply a DC bias voltage, potentially enabling tuning of mechanical properties. Furthermore, prior measurements in transmission showed device-dependent variation in lineshape under different terminations, likely due to the physical distance between devices and the

end-of-line termination being comparable to the microwave wavelength. Reflection geometry simplifies this configuration and can help mitigate such effects, particularly in future iterations that incorporate on-chip terminations, where device-termination distances are minimized and more accurately modeled.

2 Theory

2.1 Cavity Optomechanics

We follow the standard linearized Hamiltonian for a single cavity mode of frequency ω_c dispersively coupled to a mechanical resonator of frequency Ω_m through the single-photon vacuum coupling rate g_0 . The total Hamiltonian in a frame rotating at the pump frequency ω_d is

$$H = -\hbar\Delta a^\dagger a + \hbar\Omega_m b^\dagger b - \hbar g_0 a^\dagger a(b + b^\dagger) + i\hbar\sqrt{\kappa_\ell}(\alpha_{\text{in}} a^\dagger - \alpha_{\text{in}}^* a), \quad (1)$$

where $\Delta = \omega_d - \omega_c$ is the detuning and α_{in} is the coherent drive amplitude. Linearizing about the steady-state cavity field $a \rightarrow \bar{a} + d$, the linearized optomechanical coupling is $G = g_0 \bar{a}$. Throughout this work we concentrate on the “green-side” ($\Delta \simeq 0$) pumping used for thermomechanical-noise thermometry.

2.2 Input–Output Formalism and Reflection Response

For reflection measurements with an external port coupling rate κ_{ex} and internal loss rate κ_{in} , the complex reflection coefficient derived via input–output theory is

$$S_{11}(\omega) = \frac{\kappa_{\text{ex}} - \kappa_{\text{in}} + 2i(\omega - \omega_c)}{\kappa_{\text{ex}} + \kappa_{\text{in}} - 2i(\omega - \omega_c)}. \quad (2)$$

Equation (2) predicts a non-monotonic dependence of the dip depth and lineshape on κ_{in} ; increasing internal damping can either broaden or sharpen the resonance depending on the external–internal coupling ratio. We experimentally observe both regimes when toggling between open, shorted, and matched terminations.

For a mismatched termination separated by a finite length of feedline, a more complete circuit model is required. Modeling the termination as a superconducting inductor L_p magnetically coupled (mutual inductance M) to the cavity inductor L_s yields

$$S_{11} = \frac{(\omega L_p/Z_0)g - 1}{(\omega L_p/Z_0)g + 1}, \quad \text{with} \quad g = i + \frac{\kappa_{\text{ex}}}{\kappa_{\text{in}} + 2i(\omega - \omega_c)}, \quad \kappa_{\text{ex}} = \omega_c \frac{M^2}{L_p L_s}, \quad \kappa_{\text{in}} = \frac{R}{L_s}. \quad (3)$$

Equation (3) reduces to the input–output prediction (2) in the limit $\omega L_p/Z_0 \ll 1$ and serves as a convenient bridge between our circuit-level simulations and analytical fits.

2.3 Mechanical-Noise Power Spectrum and Factor Derivation relating PSD (S_x)

2.3.1 Position Spectrum from Fluctuation-Dissipation Theorem

According to the fluctuation-dissipation theorem, the position power spectral density of the thermomechanical noise in the limit $k_B T \gg \hbar\omega$ is:

$$S_{xx} = \frac{2\text{Im}[\chi]k_B T}{\omega} \quad (4)$$

where the imaginary part of the mechanical susceptibility is:

$$\text{Im}[\chi] = \frac{\Gamma_m \omega}{m[(\Omega_m^2 - \omega^2)^2 + (\Gamma_m \omega)^2]} \quad (5)$$

Therefore, the position spectral density becomes:

$$S_{xx} = \frac{2\Gamma_m k_B T}{m[(\Omega_m^2 - \omega^2)^2 + (\Gamma_m \omega)^2]} \quad (6)$$

For frequencies near the mechanical resonance ($\omega \approx \Omega_m$), this simplifies to:

$$S_{xx} = \frac{k_B T}{m\Omega_m^2} \frac{2\Gamma_m}{4(\omega - \Omega_m)^2 + \Gamma_m^2} \quad (7)$$

Converting to frequency domain ($\omega = 2\pi f$):

$$S_{xx} = \frac{1}{2} \frac{k_B T}{m\Omega_m^2} \frac{\Gamma_m/4\pi}{\pi[(f - f_m)^2 + (\Gamma_m/4\pi)^2]} \quad (8)$$

2.3.2 Equipartition Verification

Since the integral of a normalized Lorentzian is unity:

$$\int_{-\infty}^{\infty} \frac{1}{\pi} \frac{c}{(x - x_0)^2 + c^2} dx = 1 \quad (9)$$

we have:

$$\int_0^{\infty} S_{xx} df = \frac{1}{2} \frac{k_B T}{m\Omega_m^2} \quad (10)$$

Including the negative frequency contribution:

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} S_{xx} df = \frac{k_B T}{m\Omega_m^2} \quad (11)$$

in agreement with equipartition.

2.3.3 Microwave Output Spectrum

According to Eq. 2.92 of Kumar's thesis [5], the microwave output spectrum of the upper sideband of a green pump is, in units of photons:

$$S = \frac{16\Gamma_m \kappa_r g_0^2 n_d n_b}{4(\omega - \Omega_m)^2 + \Gamma_m^2} \frac{1}{\kappa_{\text{tot}}^2 + 4\Omega_m^2} \quad (12)$$

where $n_b = \frac{k_B T}{\hbar \Omega_m}$ is the thermal phonon occupation number in the high-temperature limit.

2.3.4 Factor Relation Derivation

Substituting the thermal occupation and rearranging:

$$S = \frac{4\kappa_r g_0^2 n_d n_b}{(\kappa_{\text{tot}}^2 + 4\Omega_m^2)} \frac{k_B T}{m\Omega_m^2} S_x \quad (13)$$

$$= \frac{4\kappa_r g_0^2 n_d}{\hbar m \Omega_m} \frac{1}{\kappa_{\text{tot}}^2 + 4\Omega_m^2} S_x \quad (14)$$

$$= \frac{2\kappa_r G_{\text{opto}}^2 n_d}{\kappa_{\text{tot}}^2 + 4\Omega_m^2} S_x \quad (15)$$

where S_x is the single-sided power spectral density and $G_{\text{opto}} = g_0 \sqrt{n_d}$ is the optomechanical coupling strength.

2.3.5 Intracavity Photon Number

From Eq. 2.67 of Kumar's thesis, the intracavity photon number is:

$$n_d = \frac{4\kappa_l P_{\text{RT}} G_{\text{atten}}}{\hbar \omega_d [\kappa_{\text{tot}}^2 + 4(\omega_c - \omega_d)^2]} \quad (16)$$

where P_{RT} is the pump power at our room temperature reference point, G_{atten} is the attenuation between the reference point and the sample, and $\kappa_l = \kappa_r = \Gamma_{\text{ex}}$ from our fits.

2.3.6 Device Parameters and Single-Photon Coupling

For our device parameters:

- $d = 200$ nm (air gap)
- $t = 200$ nm (drum thickness)
- $R = 100$ μm (drum radius)
- $\sigma_m = 350$ MPa (Tensile stress in aluminum)

From Eq. 47 of Youssefi et al. [1], the effective mass is:

$$m = 0.27 \rho \pi R^2 t \quad (17)$$

From Eq. 54 of the same reference, the single-photon coupling rate is:

$$g_0 = 0.37 \sqrt{\frac{\hbar \omega_c}{2d}} (R^2 t^2 \rho \sigma_m)^{-1/4} \quad (18)$$

2.3.7 Signal at Spectrum Analyzer

The signal reaching the spectrum analyzer is, in units of W/Hz = Joules:

$$S_{\text{RT}} = \hbar \omega_d G_{\text{amps}} S = A S_x \quad (19)$$

where the factor A is:

$$A = \frac{P_{\text{RT}} G 4\kappa_r g_0^2}{\hbar m \Omega_m} \frac{1}{\kappa_{\text{tot}}^2 + 4\Omega_m^2} \frac{4\kappa_l}{\kappa_{\text{tot}}^2 + 4(\omega_c - \omega_d)^2} \quad (20)$$

with $G = G_{\text{atten}} G_{\text{amps}}$ representing the total gain/attenuation factor.

2.3.8 Spectrum Analyzer Filter Analysis

The spectrum analyzer uses a Gaussian filter with RBW resolution bandwidth. We tested this (figure 8) by connecting a signal source directly to the analyzer and found that it convolves the input with:

$$f_{\text{fil}} = \exp \left[- \left(\frac{f}{\sqrt{2}\sigma_{\text{RBW}}} \right)^2 \right] \quad (21)$$

where $\sigma_{\text{RBW}} = \frac{\text{RBW}}{2\sqrt{2\ln 2}}$.

The thermomechanical spectrum output by the analyzer becomes:

$$S_{\text{an}}^{\text{RBW}} = S_{\text{RT}} \star f_{\text{fil}} \quad (22)$$

$$= A \int S_x(f') \exp \left[- \left(\frac{f - f'}{\sqrt{2}\sigma_{\text{RBW}}} \right)^2 \right] df' \quad (23)$$

This convolution can be expressed using the Voigt profile:

$$S_{\text{an}}^{\text{RBW}} = A\sqrt{2\pi} \frac{k_B T}{m\Omega_m^2} \sigma_{\text{RBW}} V[f - \Omega_m/(2\pi); \sigma_{\text{RBW}}, \Gamma_m/(4\pi)] \quad (24)$$

where $V(x; \sigma, \gamma)$ is the normalized Voigt profile.

The total integrated power is:

$$\int_{-\infty}^{\infty} S_{\text{an}}^{\text{RBW}} df = A\sqrt{2\pi} \frac{k_B T}{m\Omega_m^2} \sigma_{\text{RBW}} \quad (25)$$

This analysis allows us to extract the fundamental factor relating the microwave spectrum S to the position spectrum S_x :

$$\boxed{\frac{S}{S_x} = \frac{2\kappa_r G_{\text{opto}}^2 n_d}{\kappa_{\text{tot}}^2 + 4\Omega_m^2}} \quad (26)$$

This relationship is central to thermomechanical noise thermometry, enabling extraction of mechanical mode temperature from microwave measurements.

In the classical (high-temperature) limit $k_B T \gg \hbar\Omega_m$, the position-fluctuation spectral density of a damped harmonic oscillator is

$$S_{xx}(\omega) = \frac{2\Gamma_m k_B T}{m_{\text{eff}}} \frac{1}{(\Omega_m^2 - \omega^2)^2 + (\Gamma_m \omega)^2}, \quad (27)$$

with mechanical damping rate Γ_m and effective mass $m_{\text{eff}} = 0.27 \rho \pi R^2 t$ for a drum of radius R , thickness t , and mass density ρ .

When monitoring the upper motional sideband of a resonant ($\Delta \approx 0$) pump, input-output theory gives the microwave power spectral density at the analyzer output as

$$S_{\text{an}}(\omega) = A V\left[\frac{\omega - \Omega_m}{2\pi}; \sigma_{\text{RBW}}, \frac{\Gamma_m}{4\pi}\right], \quad A = \sqrt{2\pi} \frac{k_B T}{m_{\text{eff}} \Omega_m^2} \frac{\kappa_{\text{ex}} G^2 n_d}{\kappa_{\text{tot}}^2 + 4\Omega_m^2}, \quad (28)$$

where $V[x; \sigma, \gamma]$ denotes the Voigt profile resulting from the convolution of the Lorentzian mechanical response with the Gaussian resolution-bandwidth filter ($\sigma_{\text{RBW}} = \text{RBW}/(2\sqrt{2\ln 2})$), G is the net gain between the sample and analyzer, and n_d is the intracavity pump photon number. Equation (28) underpins the Voigt-profile fitting routine used throughout this report.

3 Experimental Setup

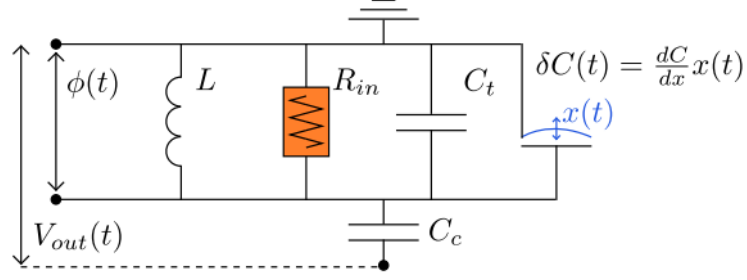


Figure 1: Electric circuit model describing the microwave cavity with a movable capacitor.

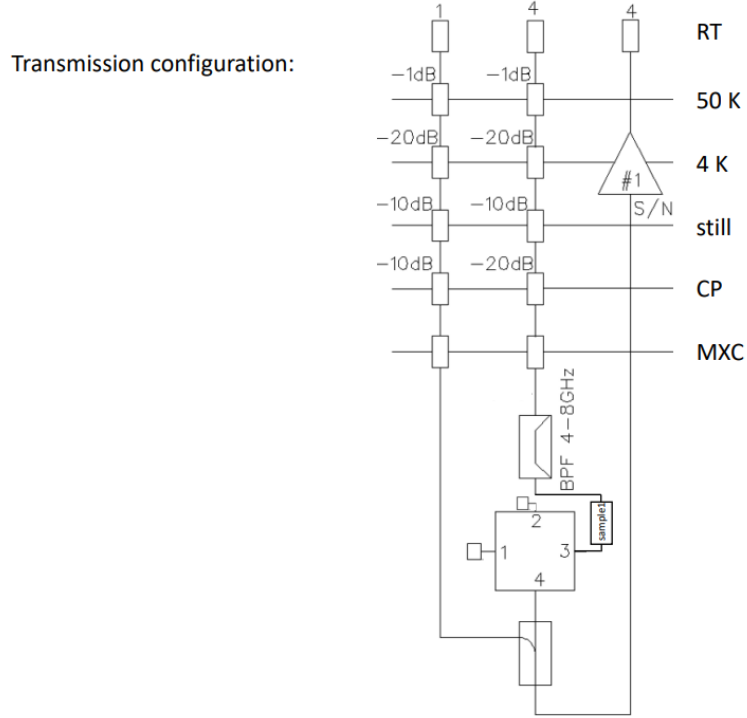


Figure 2: Cryogenic measurement circuit for transmission configuration.

All transmission-mode measurements were performed in a BlueFors dilution refrigerator at a base temperature of 12 mK. Room-temperature pump power referenced at the end of the attenuator chain is denoted P_{RT} , with a calibrated total attenuation of 66 dB. The transmitted pump tone is canceled at the output by an opposition line phase-locked to the vector-network-analyzer source, reducing the dynamic range demands on the HEMT.

For reflection geometry measurements, the same sample holder is re-wired with its feedline either (i) open-circuited, (ii) short-circuited, or (iii) $50\ \Omega$ matched at the far end. The distances between terminations and devices are comparable to the 6–8 GHz microwave wavelength, necessitating explicit modeling via Eq. (3).

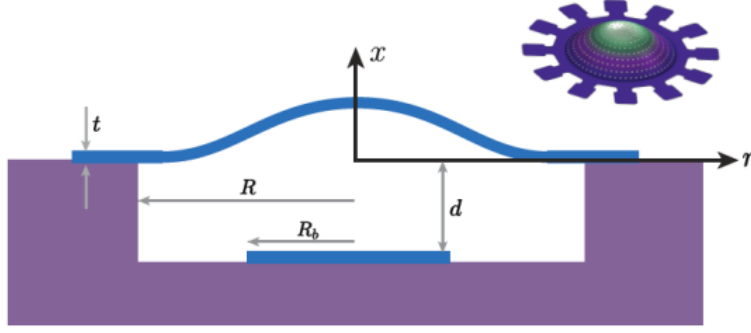


Figure 3: Optical micrograph of the drum-type nanomechanical resonator.

4 Results and Analysis

4.1 Transmission Configuration

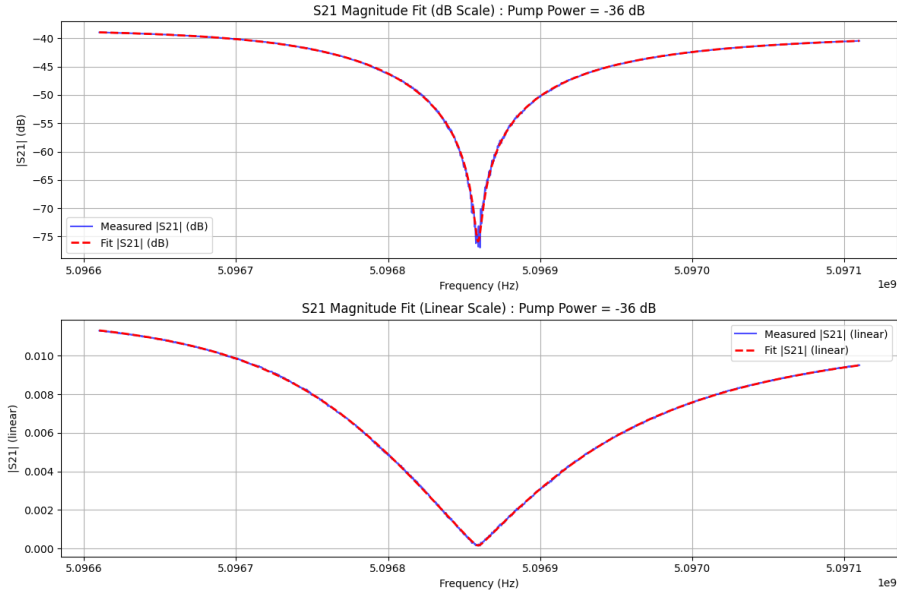


Figure 4: Representative S_{21} cavity sweep at 215 mK (blue) and best fit to the Geerlings et al. model (red).

Figure 6 shows the pump-power dependence of the internal cavity loss rate κ_{in} for the 5.1 GHz mode at three cryostat temperatures. The high-power saturation trend is consistent with two-level-system (TLS) loss quenching [3,4,5], while low-power deviations reveal additional loss channels.

4.2 Thermomechanical Noise Thermometry

Thermomechanical sideband spectra of the drum coupled to the 5.1 GHz cavity were acquired for pump powers spanning -115 to -70 dBm at the sample. Each spectrum is fit to the Voigt form of Eq. (28), yielding the mechanical linewidth Γ_m and calibrated area which, by equipartition, equals $k_B T / (m_{\text{eff}} \Omega_m^2)$ independent of pump power (Fig. 7).

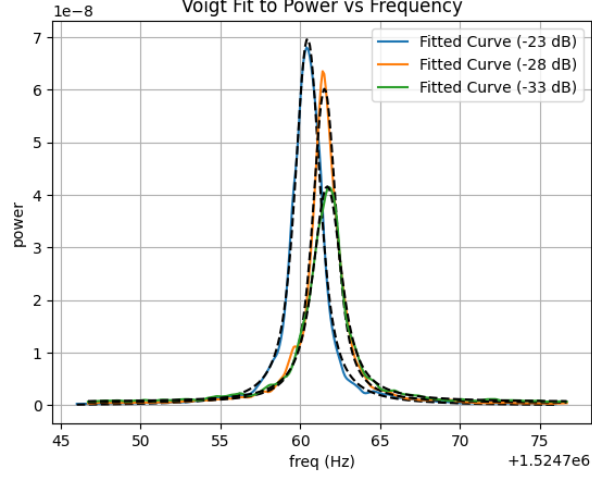


Figure 5: Voigt fit of mechanics with pump freq. 5.1 GHz

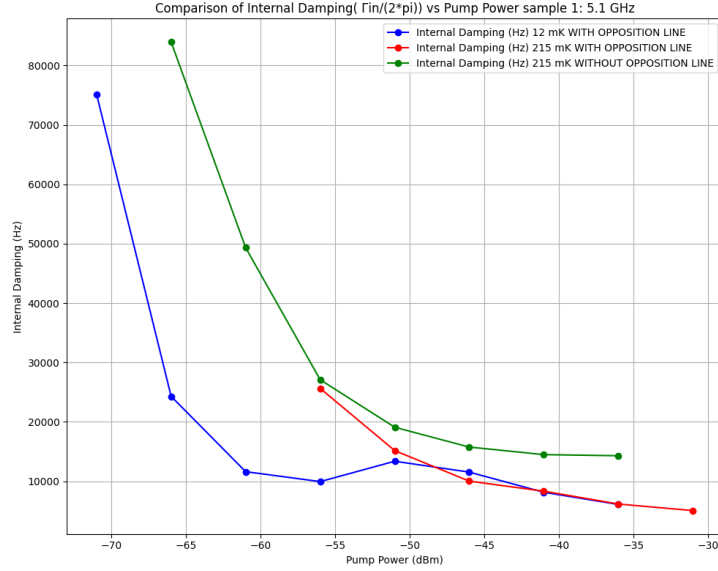


Figure 6: Internal damping κ_{in} versus Pump power for the 5.1 GHz cavity at $T = 215$ mK.

4.3 Reflection Configuration

Circuit-Model Validation. To discriminate between Kerr-type nonlinear broadening and feedline-length-induced interference, we fit the full circuit expression (3) while sweeping the effective electrical length. Best-fit parameters yield $L_p = 3.2 \pm 0.5$ nH and $M = 120 \pm 10$ pH, reproducing both the resonance depth and its non-monotonic pump-power dependence.

5 Conclusion and Outlook

Combining transmission and reflection measurements provides a comprehensive picture of microwave-loss mechanisms and mechanical-mode occupation in ultra-coherent Al drum resonators. Reflection geometry, while requiring more elaborate modeling, offers (i) reduced spurious-resonance density, (ii) compatibility with DC biasing for electrostatic tun-

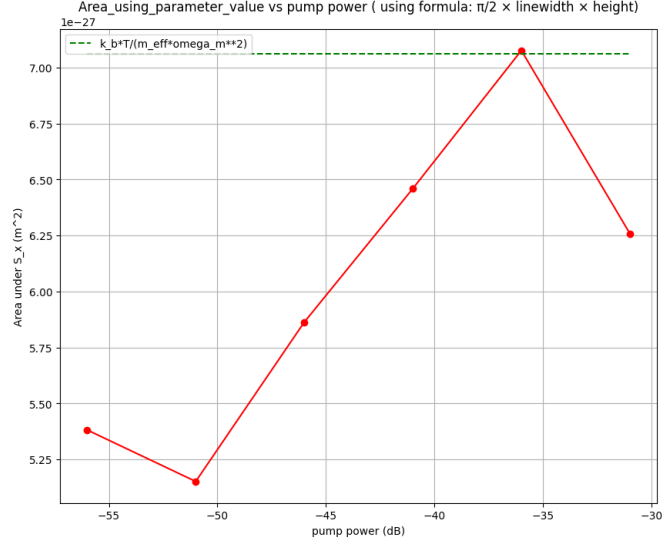


Figure 7: Area under S_x (displacement variance) versus pump power, demonstrating power-independent thermomechanical occupation for green pumping.

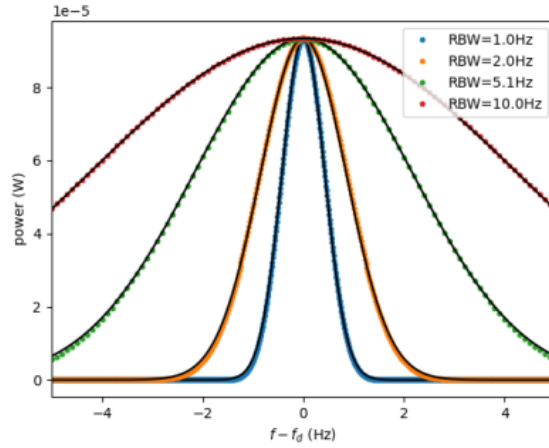


Figure 8: Fitting analyzer gaussian filter for different resolution bandwidth

ing, and (iii) direct access to S_{11} -based thermometry schemes. Immediate future work will integrate on-chip terminations to eliminate feedline-length ambiguities and extend thermomechanical noise measurements into the tens-millikelvin regime.

Appendix

.1 Data Analysis Repository

Access full analysis notebooks at:

<https://github.com/Khuspd121/Neel-Institut-CNRS-Research>

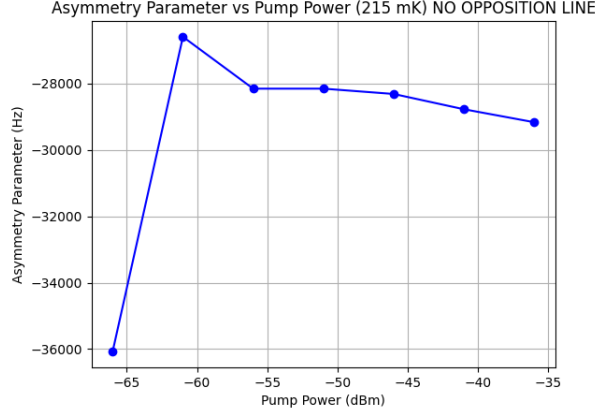


Figure 9: S_{11} Fitted asymmetry parameter versus pump power for the 5.1 GHz device at 215 mK (transmission).

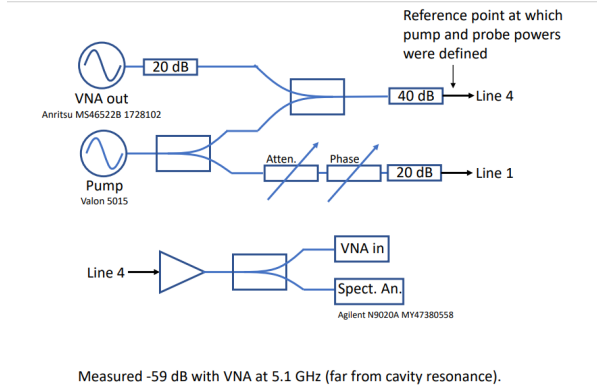


Figure 10: External circuit diagram for reflection-mode measurements.

References

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- [5] Kumar, S. *Low Temperature Microwave Optomechanics*. PhD thesis, Univ. Grenoble Alpes (2021).
- [6] Chegnizadeh, M. *et al.* Science **386**, 1383–1388 (2024).
- [7] Youssefi, A. *et al.* arXiv:2501.03211 (2025).



Figure 11: Inside the cryostat

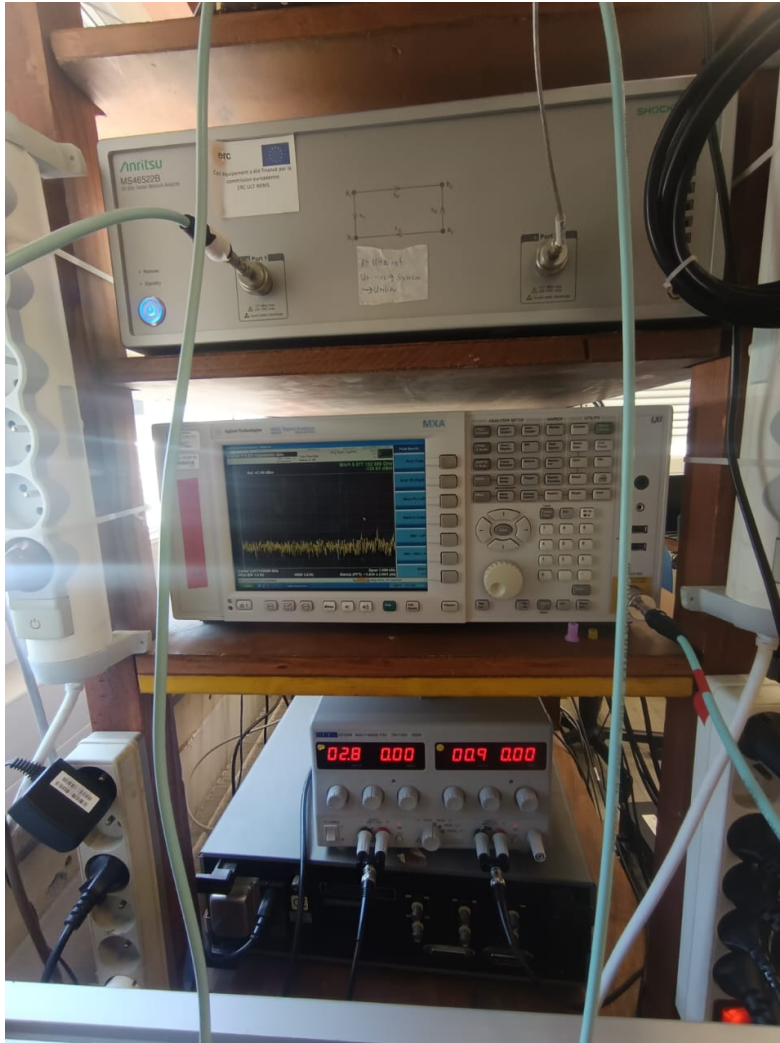


Figure 12: VNA, Spectrum Analyzer, Opposition line Amplitude & Phase Control